

FIR Filters

Digital Filters

FIR filter

- $y(n) = \sum_{k=0}^{M-1} b_k x(n-k)$
- no feedback
- Non-recursive realizations
- $H(z) = \sum_{k=0}^{M-1} b_k z^{-k}$
- All poles located at origin
i.e. stable inherently.
- Linear phase.
- for sharp magnitude response,
large memory is required
- Lesser finite word length effect
(since poles location = 0).

IIR filter

- $y(n) = -\sum_{k=1}^{N-1} a_k y(n-k) + \sum_{k=0}^{M-1} b_k x(n-k)$
- feedback
- realised recursively.
- $H(z) = \frac{\sum_{k=0}^{M-1} b_k z^{-k}}{1 + \sum_{k=1}^{N-1} a_k z^{-k}}$
- look for stability during designing.
- Non-linear phase.
- Smaller memory requirements.

Digital filters — is a discrete-time system designed to reshape spectrum of I/P signal to produce desired O/P characteristics.

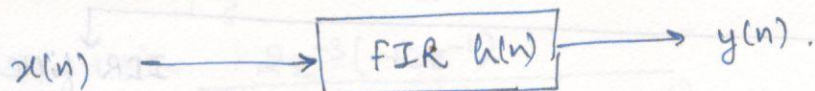
- operates with digital samples of I/P signal
- based on linear difference equations.
- consists of adders, multipliers, delay elements implemented in digital logic.

Linear & non-linear phase

- A linear phase response maintains phase relationship of the freq. components of I/P signal in pass-band.
- Non-linear phase relationship will cause phase distortion of I/P signal. ~~that~~ ~~for~~ This is because the frequency components in the signal will each be delayed by an amount not-proportional to the frequency, thereby altering their harmonic relationship.

FIR

Condition for a filter to have linear phase response.



length = M
order = M-1.

$$\text{DTFT}\{h(n)\} = H(e^{j\omega}) = |H(e^{j\omega})| e^{j\angle H(e^{j\omega})} = M(\omega) \cdot e^{j\theta(\omega)}$$

Frequency response

Mag. resp.

phase resp.

$$\angle H(e^{j\omega}) = \theta(\omega) \Rightarrow \text{Phase response}$$

(requires to be linear in passband then it only introduces delay).

Linear phase $\rightarrow$ represented by 2 types.

$$\begin{cases} \theta(\omega) = -\alpha\omega, & \Rightarrow \text{condition: } h(n) = h(M-1-n): \text{symmetric} \\ \theta(\omega) = \beta - \alpha\omega, & \Rightarrow \text{condition: } h(n) = -h(M-1-n): \text{anti-symmetric} \end{cases}$$

β, α : constant.

$\therefore$ Linear phase condition: $\{h(n) = \pm h(M-1-n)\}$.

Time delay experienced by a single frequency signal.

$$\tau_p = -\frac{\theta(\omega)}{\omega}$$

phase delay.

Time delay experienced by a signal with multiple freq. components that are not harmonically related.

$$\tau_g = -\frac{d\theta(\omega)}{d\omega}$$

group delay.

If $\tau_g(\omega)$ is const. $\Rightarrow$ all freq. components are delayed by same interval.

constant for linear phase.

Consider

$$\theta(\omega) = -\alpha\omega = -\tau\omega = \angle H(e^{j\omega})$$

Taken linear phase, checking/proving condition.

$$H(e^{j\omega}) = \sum_{n=0}^{M-1} h(n) e^{-j\omega n}$$

$$= \sum_{n=0}^{M-1} h(n) \cos \omega n - j \sum_{n=0}^{M-1} h(n) \sin \omega n$$

$$\angle H(e^{j\omega}) = \theta(\omega) = \tan^{-1} \left\{ \frac{-\sum_{n=0}^{M-1} h(n) \sin \omega(n)}{\sum_{n=0}^{M-1} h(n) \cos \omega(n)} \right\} = -\alpha \omega$$

$$= \frac{\sum_{n=0}^{M-1} h(n) \sin(\omega n)}{\sum_{n=0}^{M-1} h(n) \cos \omega n} = \frac{\sin \alpha \omega}{\cos \alpha \omega}$$

$$\Rightarrow \sum_{n=0}^{M-1} h(n) \{ \sin \omega n \cos \alpha \omega - \sin \alpha \omega \cos \omega n \} = 0$$

$$\Rightarrow \sum_{n=0}^{M-1} h(n) \sin[\omega(n-\alpha)] = 0$$

Solution of this equation

$$h(n) = h(M-1-n), \quad 0 \leq n \leq M-1$$

$$\text{and } \alpha = \frac{M-1}{2} = \tau_p = \tau$$

Ex: let $M=5$.

$$\alpha = \tau_p = \frac{M-1}{2} = 2$$

$$h(n) = h(4-n), \quad 0 \leq n \leq 4$$

$$\Rightarrow h(0) = h(4)$$

$$h(1) = h(3)$$

$$h(2) = h(2) \quad \rightarrow \text{Symmetric abt } h(2)$$

$$\text{From eq. (1)} \quad \sum_{n=0}^5 h(n) \cdot \sin(\omega(n-2)) = 0$$

$$\Rightarrow h(0) \sin(-2\omega) + h(1) \sin(-\omega) + h(2) \sin(0) + h(3) \sin(\omega) + h(4) \sin(2\omega)$$

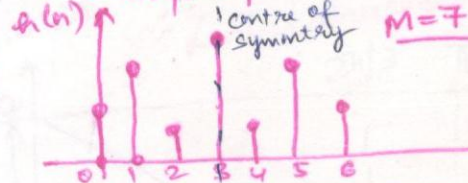
$$= 0$$

for this case of phase delay. $\tau_p = \tau_g$

① Therefore, FIR filters will have constant phase & group delays if these conditions are satisfied.

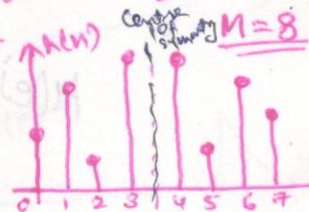
② For M -odd, $\tau = \frac{M-1}{2}$ = integer

$\therefore h(n)$ is symmetric abt sample pt. τ .



For M -even, $\tau = (M-1)/2$ = non-integer

$\therefore h(n)$ symmetric abt center b/w coefficients $\frac{M-1}{2}$ & $M/2$.



Choosing second delay.

for applications with only const. τ_g

$$D(\omega) = \beta - \alpha \omega$$

Generalized
linearity
~~filter~~

$$h(n) = -h(M-1-n)$$

$$L = \tau = \frac{M-1}{2}$$

Amplitude response.

real response

ie. can be negative
~~only~~ also.

Mag. resp.

always positive.

Frequency response of FIR filters.

$$H(e^{j\omega}) = H_r(e^{j\omega}) \cdot e^{j\theta(\omega)}$$

real $\leftarrow H_r(e^{j\omega}) \Rightarrow$ Amplitude response $\Rightarrow$ real response
can be negative also.

$\theta(\omega) \Rightarrow$ phase response.

$$D(\omega) = \beta - \alpha \omega, (\beta = 0 \text{ or } \pi/2)$$

symmetric for anti-symmetric.

example - consider

$$h(n) = \{ 1 \quad 1 \quad 1 \} \dots \text{length 3 FIR filter}$$

compute freq. response.

$$H(e^{j\omega}) = \sum_{n=0}^2 h(n) \cdot e^{-j\omega n}$$

$$= 1 + e^{-j\omega} + e^{-j2\omega}$$

$$= e^{-j\omega} (e^{j\omega} + 1 + e^{-j\omega})$$

$$= e^{-j\omega} (1 + 2\cos\omega)$$

$$H(e^{j\omega}) = (1 + 2 \cos \omega) \cdot e^{-j\omega}$$

$$= H_2(e^{j\omega}) \cdot e^{j\theta(\omega)}$$

Amplitude response: $H_2(e^{j\omega}) = 1 + 2 \cos \omega$; $\{-\pi \leq \omega \leq \pi\}$ {Amplitude response}

Phase associated with amplitude response: $\angle H(e^{j\omega}) = \theta(\omega) = -\omega$, $\{-\pi \leq \omega \leq \pi\}$ {Phase response}

This is linear (in all range).

magnitude response

$$|H(e^{j\omega})| = |1 + 2 \cos \omega|$$

$$\angle H(e^{j\omega}) = \begin{cases} -\omega & , 0 \leq \omega \leq 2\pi/3 \\ \pi - \omega & , 2\pi/3 < \omega \leq \pi \end{cases}$$

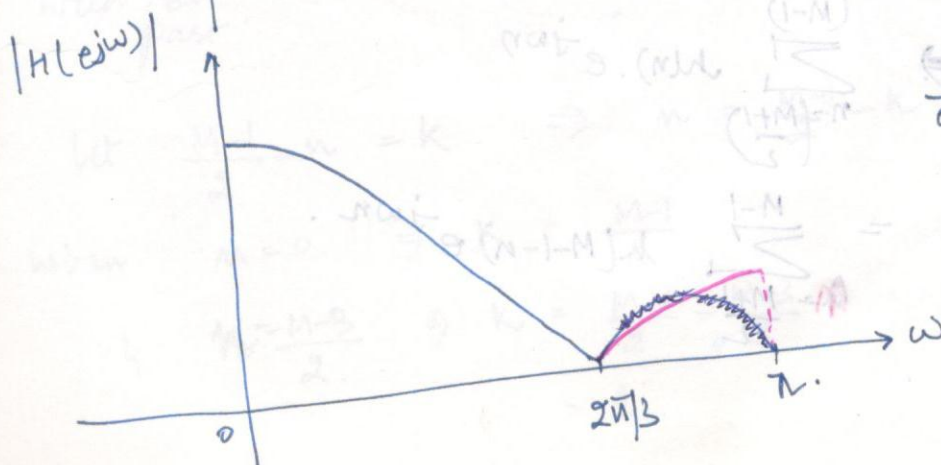
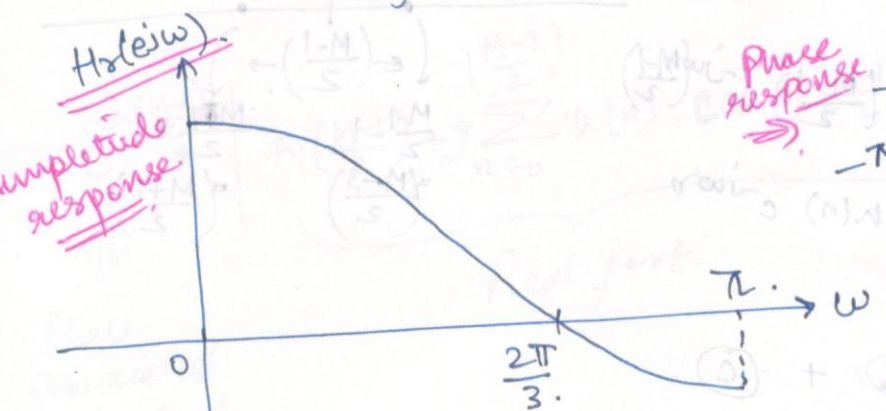
$$0 \leq \omega \leq \pi$$

This is piecewise linear -

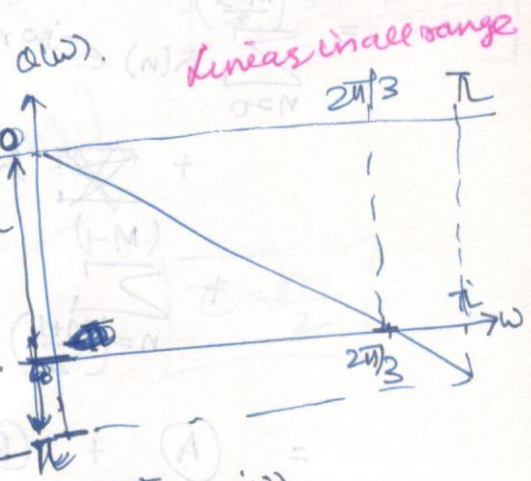
(since range is breaking in b/w).

because in this range mag. becomes negative if $\theta(\omega) = -\omega$.

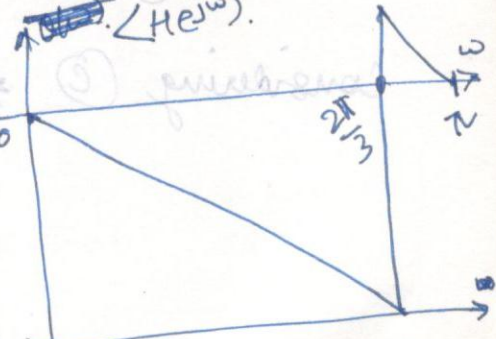
Amplitude response



Phase response



Phase response



Piecewise linear.

Frequency response of FIR filters.

$h(n) = \pm h(M-1-n)$... linear phase

Combining cases of symmetry & antisymmetry with odd & even M there are 4 types of FIR filters.

Type 1 : Symmetric impulse response.
Odd length (M-odd)
(order = M-1 ... even)

$\Rightarrow h(n) = +h(M-1-n)$ $0 \leq n \leq M-1$

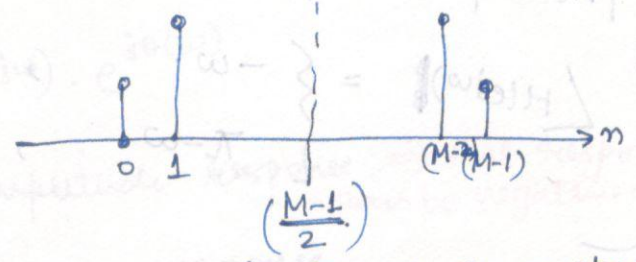
Freq. Response

$H(e^{j\omega}) = ?$

or $H(z) = ?$

Transfer functions

$\tau = \frac{M-1}{2}$



M length

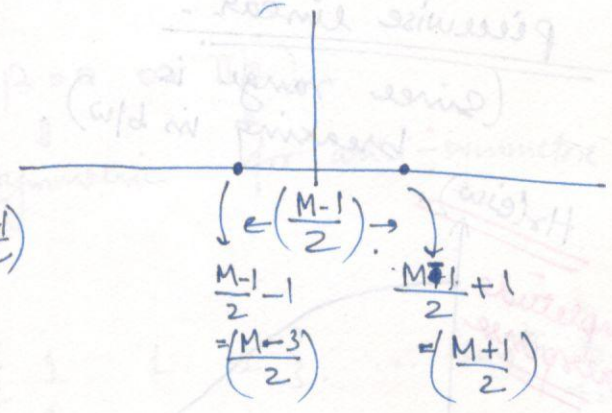
$H(e^{j\omega}) = \sum_{n=0}^{M-1} h(n) e^{-j\omega n}$

~~3 range~~ Dividing in 3 parts.

$= \sum_{n=0}^{\frac{M-3}{2}} h(n) e^{-j\omega n}$

$+ h(\frac{M-1}{2}) e^{-j\omega(\frac{M-1}{2})}$

$+ \sum_{n=\frac{M+1}{2}}^{M-1} h(n) e^{-j\omega n}$



$= \textcircled{A} + \textcircled{B} + \textcircled{C}$

Considering $\textcircled{C} \Rightarrow \sum_{n=\frac{M+1}{2}}^{M-1} h(n) e^{-j\omega n}$

$= \sum_{m=\frac{M+1}{2}}^{M-1} h(M-1-m) e^{-j\omega m}$

$$\text{Let } p = M-1-n \Rightarrow n = M-1-p$$

$$M = \frac{M+1}{2} \Rightarrow p = M-1-\frac{M+1}{2} = \frac{M-3}{2}$$

$$n = M-1 \Rightarrow p = 0$$

$$\Rightarrow \sum_{p=0}^{\frac{M-3}{2}} h(p) \cdot e^{-j\omega(M-1-p)}$$

Replacing p with n .

$$\Rightarrow \sum_{n=0}^{\frac{M-3}{2}} h(n) \cdot e^{-j\omega(M-1-n)}$$

Substituting (C)

$$H(e^{j\omega}) = \sum_{n=0}^{\frac{M-3}{2}} h(n) \left[e^{-j\omega n} + e^{-j\omega(M-1-n)} \right] + (B)$$

$$= h\left(\frac{M-1}{2}\right) e^{-j\omega\left(\frac{M-1}{2}\right)} + \sum_{n=0}^{\frac{M-3}{2}} h(n) \cdot e^{-j\omega\left(\frac{M-1}{2}\right)} \left[e^{+j\omega\left(\frac{M-1}{2}-n\right)} + e^{-j\omega\left(\frac{M-1}{2}-n\right)} \right]$$

$$= e^{-j\omega\left(\frac{M-1}{2}\right)} \left[h\left(\frac{M-1}{2}\right) + \sum_{n=0}^{\frac{M-3}{2}} h(n) \cdot \cos \omega\left(\frac{M-1}{2}-n\right) \right]$$

Real part.

$$\tau = \frac{M-1}{2}$$

Phase associated with Real part.

$$\text{Let } \frac{M-1}{2} - n = k \Rightarrow n = \frac{M-1}{2} - k$$

$$\text{when } n=0 \Rightarrow k = \frac{M-1}{2}$$

$$k = \frac{M-1}{2} - \frac{M-3}{2} \Rightarrow k = \frac{M-1}{2} - \frac{M-3}{2} = 1$$

$$\Rightarrow H(e^{j\omega}) = e^{-j\omega(\frac{M-1}{2})} \cdot \left[h(\frac{M-1}{2}) + \sum_{k=1}^{\frac{(M-1)}{2}} 2 \cdot h(\frac{M-1}{2} - k) \cdot \cos k\omega \right]$$

$$= \cancel{\sum_{k=0}^{\frac{(M-1)}{2}}} e^{-j\omega(\frac{M-1}{2})} \cdot \sum_{k=0}^{\frac{(M-1)}{2}} a(k) \cos(k\omega).$$

$$\text{where } a(k) = \begin{cases} h(\frac{M-1}{2}) & , k=0 \\ 2h(\frac{M-1}{2} - k) & , 1 \leq k \leq \frac{M-1}{2} \end{cases}$$

$$\Rightarrow H(e^{j\omega}) = e^{j\theta(\omega)} \cdot H_r(e^{j\omega}).$$

$$\theta(\omega) = -\omega \cdot \frac{M-1}{2} \Rightarrow \tau_p = \frac{-\theta(\omega)}{\omega} = \left(\frac{M-1}{2} \right)$$

$$H_r(e^{j\omega}) = \sum_{k=0}^{\frac{M-1}{2}} a(k) \cdot \cos(k\omega).$$

Type I is versatile filter as there is no constraint on zeros.

At $\omega=0$, $H(e^{j\omega}) = \frac{1}{\cos(0)}$

i.e. at low & high freq, there is no constraint on zeros.

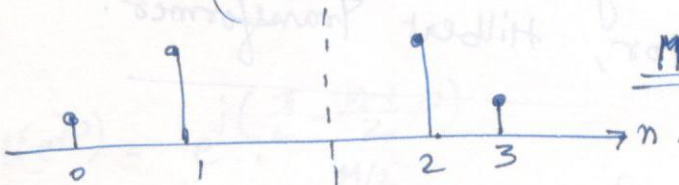
At $\omega=0$, $H(e^{j\omega})|_{\omega=0} \neq 0$.

$\omega=\pi$, $H(e^{j\omega})|_{\omega=\pi} \neq 0$.

[No disadvantage of type I as such]

Type II symmetric
length = M even.
(order = M-1 ... odd)

$h(n) = h(M-1-n)$, $0 \leq n \leq M-1$



M=4

$$H(e^{j\omega}) = e^{-j\omega \frac{M-1}{2}} \sum_{k=1}^{M/2} b(k) \cos(\omega(k-\frac{1}{2}))$$

$$b(k) = 2h(\frac{M}{2}-k), \quad 1 \leq k \leq \frac{M}{2}$$

$$H(e^{j\omega}) = e^{j\theta(\omega)} \cdot H_r(e^{j\omega})$$

$$\theta(\omega) = -\omega \left(\frac{M-1}{2} \right)$$

$$z = \frac{M-1}{2}$$

$$H_r(e^{j\omega}) = \sum_{k=1}^{M/2} b(k) \cos(\omega(k-\frac{1}{2}))$$

At $\omega=0$, $\cos(0) = 1$.

At $\omega=\pi$, $H_r(e^{j\omega})|_{\omega=\pi} = \sum_{k=1}^{M/2} b(k) \cdot \cos(\pi(k-\frac{1}{2})) = 0$.

↓

$k=1$, $\cos(\frac{\pi}{2}) = 0$.

$k=2$, $\cos(\frac{3\pi}{2}) = 0$.

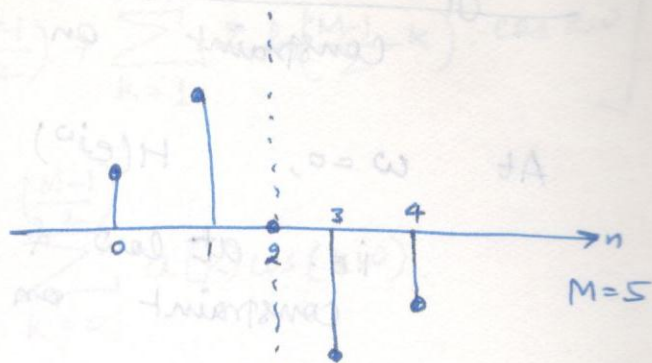
Type-3: Antisymmetric

length = M ... odd.

order = M-1 ... even.

$$h(n) = -h(M-1-n)$$

$$H(e^{j\omega}) = \sum_{n=0}^{M-1} h(n) e^{-j\omega n}$$



After derivation,

$$H(e^{j\omega}) = j e^{-j\omega(\frac{M-1}{2})} \sum_{n=0}^{\frac{(M-3)}{2}} 2h(n) \sin\left[\omega\left(\frac{M-1}{2} - n\right)\right]$$

$\left(\frac{M-1}{2}\right)$
value
at
this
pt
is always
zero.

i.e. Freq. resp is imaginary of type-3 filter.

* Type -3 is suitable for those systems whose FR is imaginary.
e.g. differentiator, Hilbert Transformer.

$$j = e^{j\pi/2}$$

$$\Rightarrow H(e^{j\omega}) = e^{j\left(\frac{\pi}{2} - \frac{M-1}{2}\omega\right)} \sum_{k=1}^{\frac{M-1}{2}} c(k) \cdot \sin(k\omega)$$

$$= e^{j\theta(\omega)} \cdot H_x(e^{j\omega})$$

← where

$$c(k) = 2h\left(\frac{M-1}{2} - k\right), \quad 1 \leq k \leq \frac{M-1}{2}$$

$$\theta(\omega) = \frac{\pi}{2} - \frac{M-1}{2}\omega = \beta - \alpha\omega$$

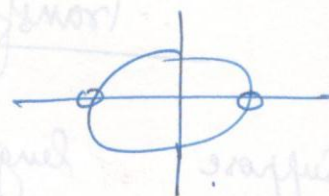
$$= \beta - \tau\omega \quad (\text{due to linear phase})$$

$$H_x(e^{j\omega}) = \sum_{k=1}^{\frac{M-1}{2}} c(k) \cdot \sin(k\omega)$$

At $\omega = 0$, $\sin(0) = 0$.

$H_z(e^{j\omega})|_{\omega=0} = 0 \quad (z=1)$

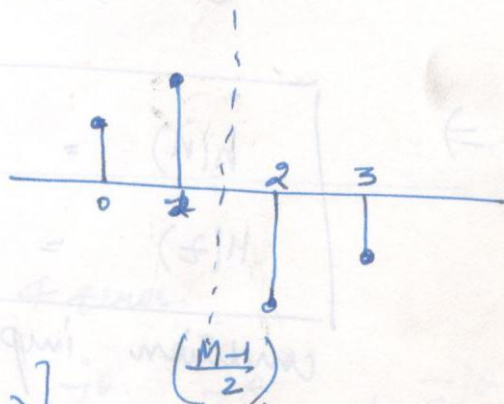
$H_z(e^{j\omega})|_{\omega=\pi} = 0 \quad (z=-1)$



Type - 3 is not suitable for LP, HP, BR.
Suitable for BP, differentiators, Hilbert transformers.

Type: 4

Antisymmetric
(length = M... even)
(order = M-1... odd)



$$H(e^{j\omega}) = e^{j(\frac{\pi}{2} - \frac{M-1}{2}\omega)} \sum_{k=1}^{M/2} d(k) \cdot \sin\left[\omega\left(k - \frac{1}{2}\right)\right]$$

$$= e^{j\theta(\omega)} H_z(e^{j\omega})$$

$$H_z(e^{j\omega}) = \sum_{k=1}^{M/2} d(k) \cdot \sin\left[\omega\left(k - \frac{1}{2}\right)\right]$$

$H_z(e^{j\omega})|_{\omega=0} = 0 \quad (z=1)$



$H_z(e^{j\omega})|_{\omega=\pi} \neq 0$

not suitable for LP & BR.

Suitable for BP, HP, diff, Hilbert Trans.

Location of Zeros of Linear-phase FIR Transfer Functions.

Suppose length ... M
order ... $M-1$

Linear phase, may be symmetric/anti-symmetric.

if $h(n) \xleftrightarrow{ZT} H(z)$.

$$h(n+M-1) \longleftrightarrow z^{M-1} \cdot H(z).$$

$$h(M-1-n) \longleftrightarrow z^{-M+1} \cdot H(z^{-1}).$$

$$\Rightarrow \begin{cases} h(n) = \pm h(M-1-n), \\ H(z) = \pm z^{-(M-1)} \cdot H\left(\frac{1}{z}\right). \end{cases}$$

condition imposed on T/F of linear phase FIR filter.

* If $z = z_0$ is a zero of $H(z)$, i.e. $H(z_0) = 0$.

$$H(z_0) = \pm z_0^{-(M-1)} H\left(\frac{1}{z_0}\right) = 0.$$

$$\Rightarrow H\left(\frac{1}{z_0}\right) = 0.$$

$\Rightarrow$ If z_0 is zero, then $\frac{1}{z_0}$ is zero.

$(z_0, \frac{1}{z_0}) \Rightarrow$ pair of zeros.

For a real coefficients filter.

$$\Rightarrow h(n) = h^*(n).$$

$$H(z) = H^*(z^*).$$

$$h(n) = \pm h(M-1-n) = h^*(n) = \pm h^*(M-1-n).$$

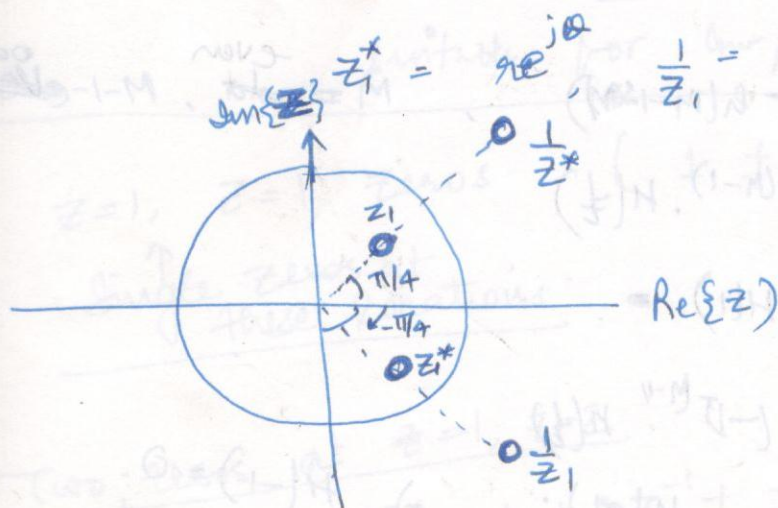
$$H(z) = \pm z^{-(M-1)} H\left(\frac{1}{z}\right) = H^*\left(\frac{1}{z^*}\right) = \pm \left(\frac{1}{z^*}\right)^{-(M-1)} H^*\left(\frac{1}{z^*}\right).$$

Let $H(z_0) = 0$, $z_0 = \text{zeros}$.

Zeros: $\left(z_0, \frac{1}{z_0}, z_0^*, \frac{1}{z_0^*} \right)$.

Suppose, zero: $z_1 = re^{j\theta}$.
 & filter: real, ie. 4 zeros.

$$z_1^* = re^{-j\theta}, \quad \frac{1}{z_1} = \frac{1}{r} e^{-j\theta}, \quad z_1^* = \frac{1}{r} e^{j\theta}.$$

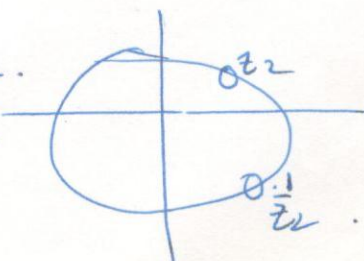


Let $z_1 = 0.5 e^{j\pi/4}$.

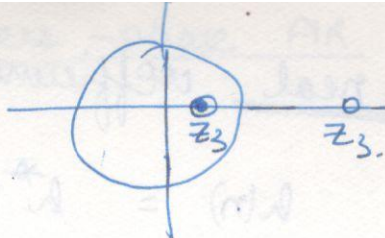
Zeros form a
mirror image
polynomial.

Suppose, $z_2 = e^{j\theta} = \frac{1}{z_2^*}$
 $z_2^* = e^{-j\theta} = \frac{1}{z_2}$

Zeros at unit circle.
(oscillators).



$$\Rightarrow z_3 = \frac{1}{z_3} = z_3^* \\ \frac{1}{z_3} = \frac{1}{z_3} = \frac{1}{z_3^*}$$



$$\begin{aligned} \Rightarrow z_4 &= 1 \\ \Rightarrow z_5 &= -1 \end{aligned} \left. \vphantom{\begin{aligned} \Rightarrow z_4 &= 1 \\ \Rightarrow z_5 &= -1 \end{aligned}} \right\} \text{Single zero only.}$$

Zeros of different Type filters.

Type I.

$$h(n) = h(M-1-n).$$

$M \dots \text{odd.}$

$$H(z) = z^{-(M-1)} H(z).$$

$$H(1) \neq 0, \quad H(-1) \neq 0.$$

Type II.

$$h(n) = -h(M-1-n)$$

$M = \overset{\text{even}}{\text{odd}}, \quad M-1 = \overset{\text{odd}}{\text{even}}.$

$$H(z) = -z^{-(M-1)} H(z)$$

$$H(1) = H(1) = 0$$

$$H(-1) = (-1)^{M-1} H(-1)$$

$$H(-1) = -H(-1)$$

$$\Rightarrow H(-1) = 0.$$

Inherently,
there are
zeros at
 $z = -1$.

Type 3:

$$h(n) = -h(M-1-n).$$

M - odd
(M-1) - even.

$$H(z) = 1 - \cancel{z}^{-(M-1)} \cdot H\left(\frac{1}{z}\right).$$

$$H(1) = -H(1) \Rightarrow \underline{H(1) = 0}.$$

$$H(-1) = H(-1) \Rightarrow \cancel{H(-1)} \cdot H(-1) = 0.$$

Two zeros

$$\underline{\underline{z=1}} \\ \underline{\underline{z=-1}}.$$

Type 4:

$$h(n) = h(M-1-n)$$

M - even
(M-1) ... odd

$$H(z) = z^{-(M-1)} \cdot H(z).$$

$$H(1) = 0.$$

$$H(-1) = H(-1).$$

Zero at $z = -1$.

substitute for $h(n)$:

$$z=1, z=1 \text{ zeros } (1 \pm z^{-1}).$$

Single zeros at these locations:

Two zeros at $z=1, z=-1$.

$$(1 \pm z^{-1})^2 = 1 \pm 2z^{-1} + z^{-2}.$$

$$\{1, \pm 2, 1\} \rightarrow \text{symmetric}.$$

Shows linear phase

* Type I FIR filter have either even number of zeros at $z=1$, ~~$z=1$~~ $z=-1$, or have no zeros. (for linear phase).

* Type II FIR filter. may have either no zero or an even no. of zeros at $z=1$, and odd number of zeros at $z=-1$

* Type -3, odd no. of zeros at $z=1$ & $z=-1$

* Type -4, odd no. of zeros at $z=1$ and no or even no. of zeros at $z=-1$

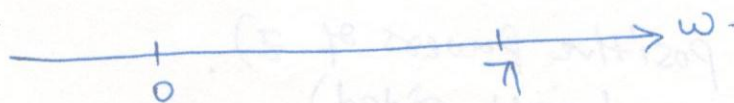
* If coefficients of all powers of z are positive, it's LPF.

* If coeff. of odd powers of z are negative, it's HPF.

FILTER SPECIFICATIONS.

Linear scale { Gain : $G = |H(e^{j\omega})|$
 Attenuation : $A = \frac{1}{|H(e^{j\omega})|} = \frac{1}{G}$

dB scale { Gain : $G_{dB} = 20 \log_{10} |H(e^{j\omega})| = 20 \log_{10} G$
 Attenuation : $A_{dB} = 20 \log_{10} \frac{1}{|H(e^{j\omega})|} = -20 \log_{10} G$
 $\Rightarrow \boxed{A_{dB} = -G_{dB}}$



ω_p ... passband edge frequency.

ω_s ... stopband edge frequency.

$[0, \omega_p]$... passband } LPF
 $[\omega_s, \pi]$... stopband }
 $[0, \omega_s]$... stopband } HPF
 $[\omega_p, \pi]$... passband }

$$\omega_p > \omega_s$$

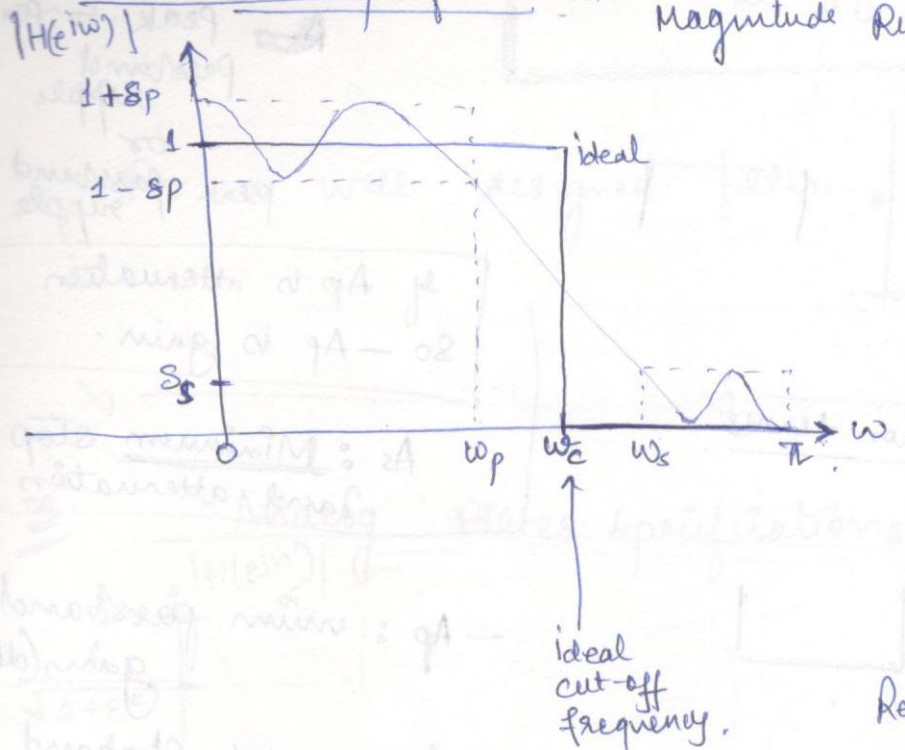
$$\omega_s > \omega_p$$

Filter specifications

16/10/14 / Tuesday

1. Absolute Specifications

Magnitude Response.



Designing filter with limited / finite coeff., oscillations occurs in passband & stopband

S_p : passband (ripples) tolerance.

S_s : stopband (ripple) tolerance.

Response oscillates b/w 0 to S_s in stopband since we plotting magnitude Resp. (Abs).

$$1 - S_p \leq |H(e^{jw})| \leq 1 + S_p, \quad 0 \leq w \leq w_p, \quad S_p \ll 1$$

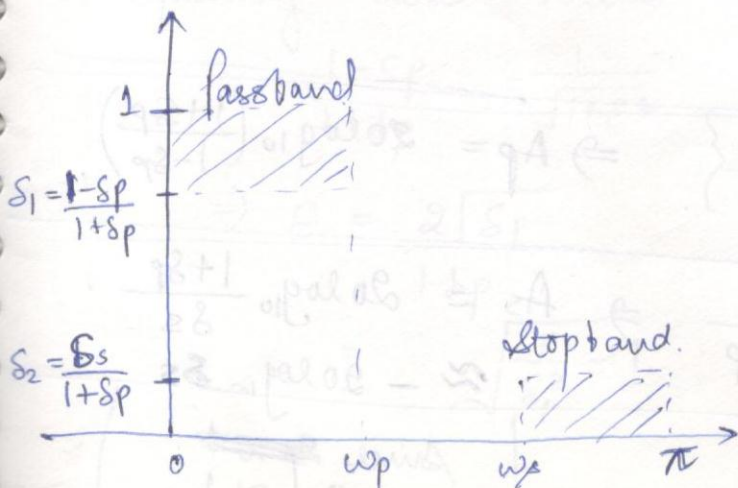
$$|H(e^{jw})| \leq S_s, \quad w_s \leq w \leq \pi, \quad S_s \ll 1$$

2. Relative Specifications

Divide all values with max value of Mag. ~~of~~ Resp. i.e. $1 + S_p$.

so also called Normalized specifications

$$|H(e^{jw})|_{\max} = 1 + S_p$$

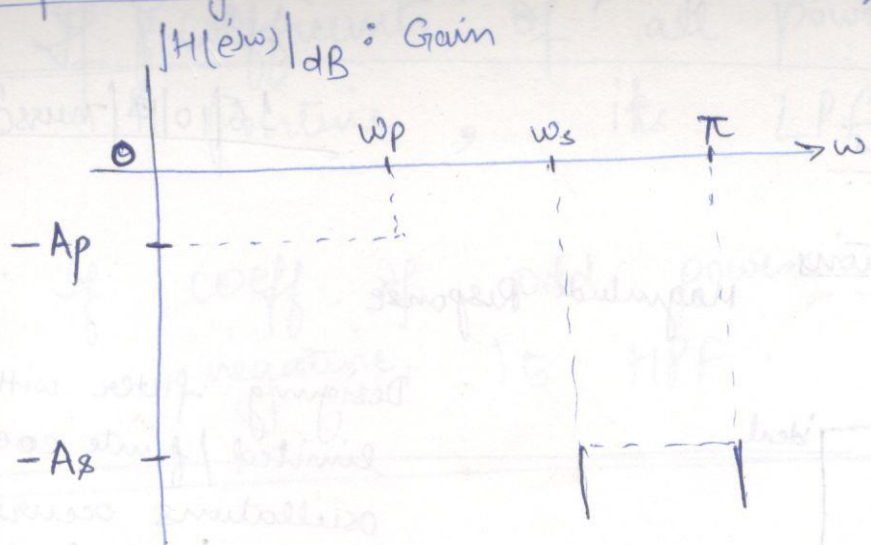


$$\frac{1 - S_p}{1 + S_p} \leq |H(e^{jw})| \leq 1, \quad 0 \leq w \leq w_p$$

$$|H(e^{jw})| \leq \frac{S_s}{1 + S_p}, \quad w_s \leq w \leq \pi$$

$$S_1 = \frac{1 - S_p}{1 + S_p}, \quad S_2 = \frac{S_s}{1 + S_p}$$

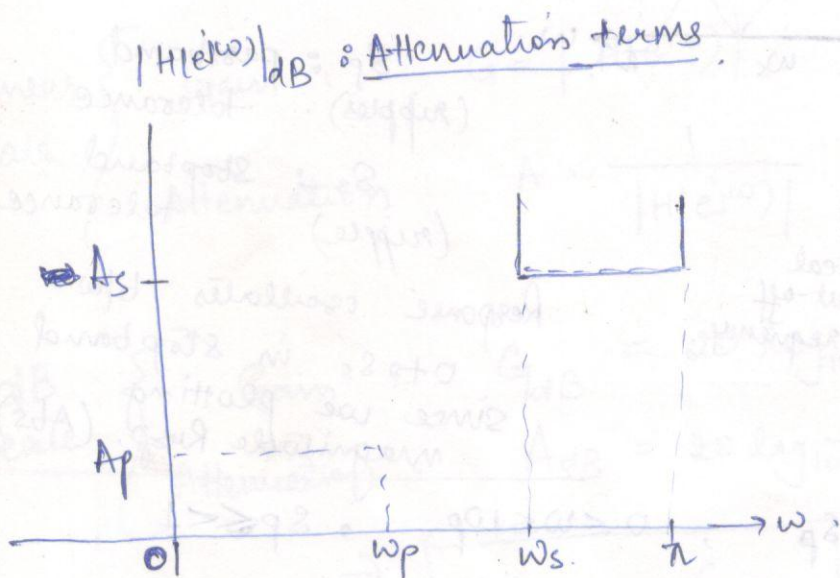
Representing this in dB ~~terms~~ in ~~terms~~ of Gain



$A_p : \text{Maximum passband attenuation or peak-to-peak passband ripple or peak passband ripple}$

If A_p is attenuation so $-A_p$ is gain.

$A_s : \text{Minimum stopband attenuation}$



$-A_p : \text{min passband gain (dB)}$
 $-A_s : \text{max stopband gain (dB)}$

$$0 \leq |H(e^{jw})|_{dB} \leq A_p, \quad 0 \leq w \leq w_p$$

$$|H(e^{jw})|_{dB} \geq A_s, \quad w_s \leq w \leq \pi$$

$$-A_p = 20 \log_{10} \left\{ \frac{1-s_p}{1+s_p} \right\} \Rightarrow A_p = 20 \log_{10} \left(\frac{1+s_p}{1-s_p} \right)$$

$$-A_s = 20 \log_{10} \frac{s_s}{1-s_p} \Rightarrow A_s = 20 \log_{10} \frac{1+s_p}{s_s} \approx -20 \log_{10} s_s$$

Since $s_p \approx 1$
 $1+s_p \approx 2$

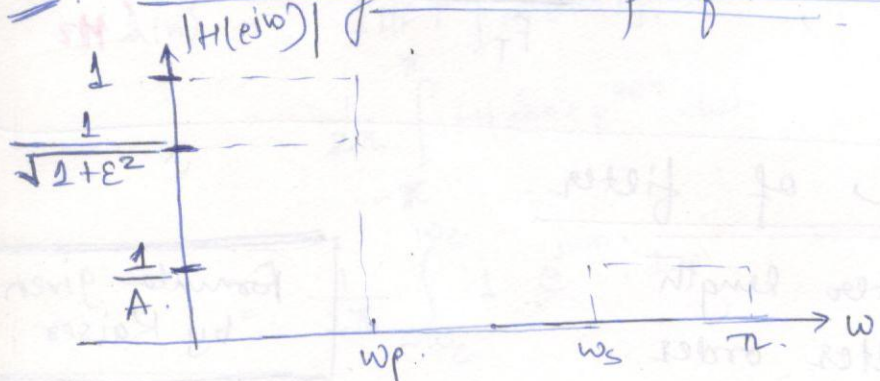
Also-

$$S_p = \frac{10^{A_p/20} - 1}{10^{A_p/20} + 1}$$

$$\delta_s = (1 + S_p) \cdot 10^{-A_s/20} \\ \approx 10^{-A_s/20}$$

for a well designed filter, $A_p \approx 0$, $A_s \gg 1$ $\rightarrow$ very high.

3. Analog Filter Specifications.



In analog filter we prefer squared mag. response.

ϵ - ripple parameter for Butterworth, $\epsilon = 1$.

$$1 \leq |H(e^jw)| \leq \frac{1}{\sqrt{1+\epsilon^2}}, \quad 0 \leq w \leq w_p.$$

$$|H(e^jw)| \leq \frac{1}{A}, \quad w_s \leq w \leq \pi.$$

Equating these values with relative specifications.

$$\frac{1 - S_p}{1 + S_p} = \frac{1}{\sqrt{1 + \epsilon^2}}$$

$$\frac{\delta_s}{1 + S_p} = \frac{1}{A}$$

$$\Rightarrow \epsilon = \frac{2\sqrt{S_p}}{1 - S_p}$$

$$A = \frac{1 + S_p}{S_p} = \frac{1}{S_2}$$

$$= \sqrt{\frac{1}{S_1^2} - 1}$$

$$A_p = -20 \log_{10} \frac{1-s_p}{1+s_p},$$

$$= -20 \log_{10} \frac{1}{\sqrt{1+\epsilon^2}}.$$

$$\Rightarrow \begin{cases} A_p = 20 \log_{10}(\sqrt{1+\epsilon^2}) \\ \epsilon = \sqrt{10^{A_p/20} - 1} \end{cases}$$

$$A_s = -20 \log_{10} \frac{s_s}{1+s_p}.$$

$$= -20 \log_{10} \left(\frac{1}{A} \right)$$

$$\begin{cases} A_s = 20 \log_{10}(A) \\ A = 10^{A_s/20} \end{cases}$$

To design digital filter with given analog specifications we need to convert analog f into digital f using relation:

$$\omega_p = \frac{\Omega_{\text{pass}}}{F_T}$$

$$\omega_s = \frac{\Omega_{\text{stop}}}{F_T}$$

F_T - Sampling
freq. in
Hz.

Estimation of order of filter.

M = filter length

$M-1$ = filter order.

Formula given
by Kaiser

length, $M = \frac{-20 \log_{10}(\sqrt{\delta_p \delta_s}) - 13}{14.6 (\omega_s - \omega_p)/2\pi} + 1.$

∴ order, $(M-1) = \left[\quad \quad \quad \right]$

Windowing Method.

$h_d(n)$... desired impulse response (infinite length).
 $w(n)$... window (length = M) finite length sequence.

$h(n) = h_d(n) \cdot w(n)$... FIR filter impulse response.

$$H(e^{j\omega}) = \frac{1}{2\pi} [H_d(e^{j\omega}) * W(e^{j\omega})]$$

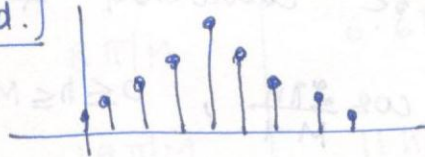
1. Rectangular $w_R(n) = \begin{cases} 1, & 0 \leq n \leq M-1 \\ 0, & \text{otherwise} \end{cases}$
2. Triangular window (Bartlett).

$$w_T(n) = \begin{cases} 1 - \frac{2|n - \frac{M-1}{2}|}{M-1}, & 0 \leq n \leq M-1 \\ 0, & \text{otherwise} \end{cases}$$

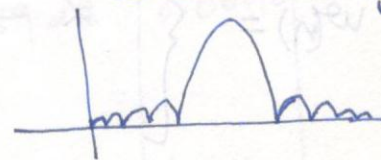
Gradual Transition
 i.e. ~~not~~ Tapered.

amp.
 gradually
 decreases.

$$w_T(n)$$

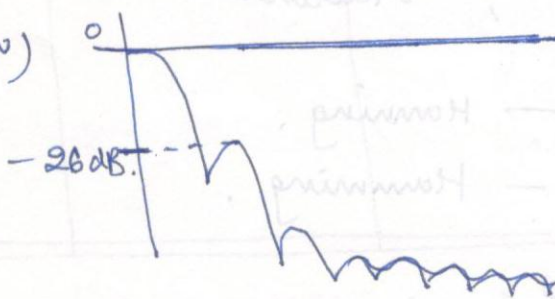


DTFT is sine square.



$$W_T(e^{j\omega})$$

(dB)



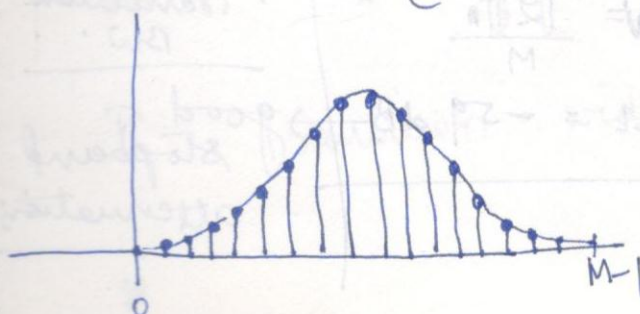
$$MLW = \frac{8\pi}{M}$$

$$RSL = -26dB$$

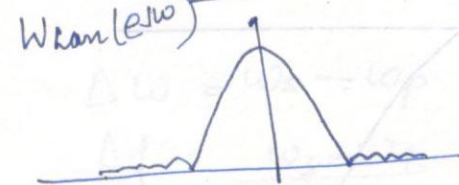
3. Hanning (Hann) window.

$$w_{Hann}(n) = \begin{cases} 0.5 - 0.5 \cos \frac{2\pi n}{M-1}, & 0 \leq n \leq M-1 \\ 0, & \text{otherwise} \end{cases}$$

Low MLW is desirable and is least in Rectangular



$$W_{Hann}(e^{j\omega})$$



Very low side lobes.

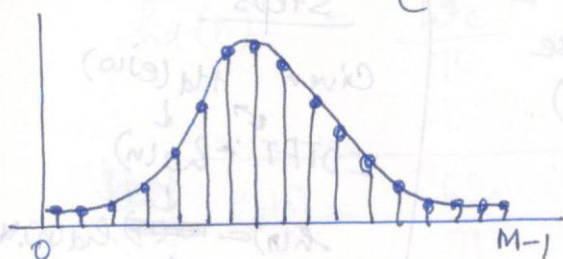
$$MLB = \frac{8\pi}{M}$$

$$RSL$$

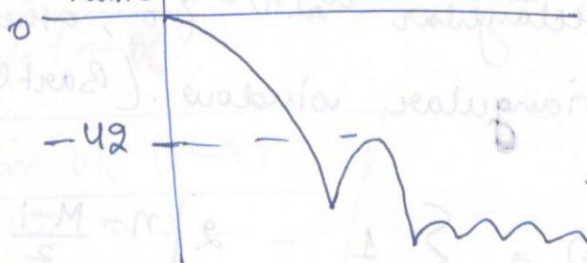
4. Hamming Window

$$w_{\text{ham}}(n) = \begin{cases} 0.54 - 0.46 \cos \frac{2\pi n}{M-1}, & 0 \leq n \leq M-1 \\ 0, & \text{otherwise} \end{cases}$$

$w_{\text{ham}}(n)$



$W_{\text{ham}}(e^{j\omega}) \text{ dB}$



$$MLW = \frac{8\pi}{M}, \quad RSL = -42 \text{ dB}$$

Above 2 windows can be combined & written as

$$w(n) = \begin{cases} \alpha - (1-\alpha) \cos \frac{2\pi n}{M-1}, & 0 \leq n \leq M-1 \\ 0, & \text{otherwise} \end{cases}$$

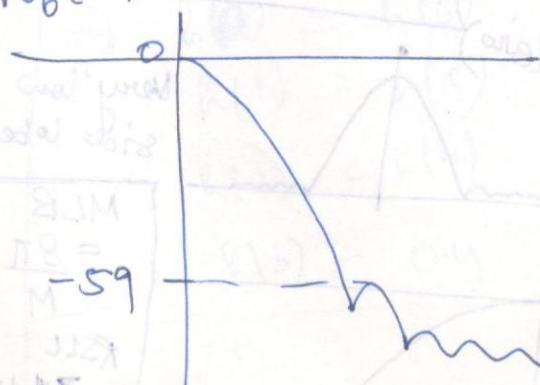
where, $\alpha = 0.5$ — Hanning

$\alpha = 0.54$ — Hamming

5. Blackman Window

$$w_b(n) = \begin{cases} 0.42 - 0.5 \cos \frac{2\pi n}{M-1} + 0.08 \cos \frac{4\pi n}{M-1}, & 0 \leq n \leq M-1 \\ 0, & \text{otherwise} \end{cases}$$

$W_b(e^{j\omega}) \text{ dB}$



$$MLW = \frac{12\pi}{M}$$

$$RSL = -59 \text{ dB}$$

not good
Transition
BW

good
stopband
attenuation

Filter Design Steps using Windowing Method

Given Specifications : $\omega_p, \omega_s, A_p, A_s$.
or
 $H_d(e^{j\omega})$.



For Design of LFF. with $\omega_c = \pi/2$, $M=41$

Window:	Main Lobe width	Exact Transition width ($\Delta\omega$)	Relative Sidelobe Level	$S_p = S_s$	Max. Awp. or Aps (dB)	Min Awp. or Aps (dB)
Rectangular	$\frac{4\pi}{M}$	$\frac{1.8\pi}{M}$	-13 dB	0.095	1.6553	21
Triangular	$\frac{8\pi}{M}$	$\frac{6.1\pi}{M}$	-26 dB	0.051	0.8861	26
Hanning	$8\pi/M$	$6.2\pi/M$	-31 dB	0.0063	0.1094	44
Hamming	$8\pi/M$	$6.6\pi/M$	-42 dB	0.0019	0.0330	54.5
Blackman	$12\pi/M$	$11\pi/M$	-59 dB	0.00018	0.0031	75

Step I. Determine ~~$H_d(n)$~~ $H_d(n)$ from $H_d(e^{j\omega})$.

$$H_d(e^{j\omega}) = \begin{cases} 1 & |\omega| \leq \omega_c \\ 0 & \omega_c < |\omega| \leq \pi \end{cases}$$

$$\omega_c = \frac{1}{2} [\omega_s + \omega_p]$$

II Transition width : $\Delta\omega = \omega_s - \omega_p$
 $\Delta f = \frac{\omega_s - \omega_p}{2\pi}$

From A_{pass} & A_{stop} , Calculate S_p & S_s

$$S_p = \frac{10^{A_p/20} - 1}{10^{A_p/20} + 1}, \quad S_s = 10^{-A_s/20}$$

$$S = \min[S_p, S_s]$$

IV. According to S , compute attenuation, A .

$$A = -20 \log_{10} S.$$

actual stop band attenuation.

V. From this table, look for A_{ws} (or A_{stop}), which is greater than A , calculated above.

So according to given specifications, we are choosing window which should be used for our problem.

for $A_{\text{ws}} > A$, options select for window to be used.

$$VI. \Delta\omega = \frac{C}{M} \Rightarrow M = \frac{C}{\Delta\omega}$$

C = upper term in $\Delta\omega$.

compute length of

window, M .

If M = even, choose ~~next~~ higher odd number for

$$VII. h(n) = h_a(n) \cdot w(n)$$

$$\text{general } H_d(e^{j\omega}) = \begin{cases} e^{-j\omega} & , \quad |\omega| \leq \omega_c \\ 0 & , \quad \omega_c < |\omega| \leq \pi. \end{cases}$$

if $M=22$, can choose 25 versatile type I filter.

Ques.

Given $H_d(e^{j\omega}) = \begin{cases} e^{-j3\omega}, & 0 \leq \omega \leq \pi/2 \\ 0, & \pi/2 < \omega \leq \pi \end{cases}$ $-\omega_c \leq \omega \leq \omega_c$
 $M=7, \omega_c = 1 \text{ rad/sample.}$

$\omega_c(n) : \text{MLW} = 8\pi/M \text{ \& RSL} = -31 \text{ dB.}$

Soln

Step 1 :

$\omega_c = 1 \text{ rad/sample.}$

$\tau = 3$

also $\tau = \frac{M-1}{2} \Rightarrow M=7.$

$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\omega}) \cdot e^{j\omega n} \cdot d\omega$

$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} (e^{-j3\omega}) \cdot e^{j\omega n} \cdot d\omega$

$= \frac{1}{2\pi} \left(\frac{e^{j\omega(n-3)}}{j(n-3)} \right) \Big|_{-\omega_c}^{\omega_c} = \frac{1}{\pi(n-3)} \left(\frac{e^{j\omega_c(n-3)} - e^{-j\omega_c(n-3)}}{2j} \right)$

$h_d(n) = \frac{\sin \omega_c(n-3)}{\pi(n-3)} = \frac{\sin(n-3)}{\pi(n-3)}$: Symmetric abt $n=3.$

$h_d(3) = \frac{d}{dn} \frac{\sin(n-3)}{\pi(n-3)} \Big|_{n=3} = \frac{\cos(n-3)}{\pi} \Big|_{n=3}$ But at $n=3, h_d(3) = 0/0$
 $= \frac{1}{\pi} = 0.3183$ $\therefore$ apply L'Hospital Rule

$h_d(0) = h_d(6) = 0.015$

$h_d(1) = h_d(5) = 0.1447$

$h_d(2) = h_d(4) = 0.2678$

Step 2 : Given $MLW = \frac{8\pi}{M}$ & $RSLL = -31 \text{ dB}$

Suitable window: Hanning Window.

$$w(n) = \left\{ 0.5 - 0.5 \cos\left(\frac{\pi n}{3}\right), 0 \leq n \leq 6 \right\} = \left\{ \underset{\uparrow}{0}, 0.25, 0.75, 1, 0.75, 0.25, 0 \right\}$$

Step 3 : Multiply desired $h_d(n)$ with $w(n)$ to get Causal FIR filter impulse response, $h(n)$.

$$\begin{aligned} \Rightarrow h(n) &= h_d(n) \cdot w(n) \\ &= \{ 0.015 \times 0, 0.1447 \times 0.25, \dots \} \\ &= \left\{ \underset{\uparrow}{0}, 0.0362, 0.2009, 0.3183, 0.2009, \right. \\ &\quad \left. 0.0362, 0 \right\} \end{aligned}$$

$$\Rightarrow h(n) = h(M-1-n) = h(6-n)$$

Step 4 : Frequency response of designed FIR filters:

$$H(e^{j\omega}) = \sum_{n=0}^6 h(n) e^{j\omega n}$$

$$= h(0) + h(1)e^{j\omega} + h(2)e^{j2\omega} + h(3)e^{j3\omega} + h(4)e^{j4\omega} + h(5)e^{j5\omega} + h(6)e^{j6\omega}$$

$$= e^{-j3\omega} [h(0)e^{j3\omega} + h(1)e^{j2\omega} + h(2)e^{j\omega} + h(3) + h(4)e^{-j\omega} + h(5)e^{-j2\omega} + h(6)e^{-j3\omega}]$$

$$= e^{-j3\omega} [h(3) + h(2)[e^{j\omega} + e^{-j\omega}] + h(1)[e^{j2\omega} + e^{-j2\omega}] + h(0)[e^{j3\omega} + e^{-j3\omega}]]$$

$$H(e^{j\omega}) = \underbrace{e^{-j3\omega}}_{\text{Phase}} \left[\underbrace{0.3183 + 0.4018 \cos \omega + 0.0724 \cos 2\omega}_{\text{Mag}} \right]$$

Q. Design linear phase - LP digital filter

$$H_d(e^{j\omega}) = \begin{cases} e^{-j3\omega} & 0 \leq |\omega| \leq \pi/2 \\ 0 & \pi/2 < |\omega| \leq \pi \end{cases}$$

using Bartlett window, choosing suitable M (filter length)

Find impulse & frequency response.

Determine system function & difference eqn.

Draw linear phase structure of filter.

Solⁿ.

given $\omega_c = \frac{\pi}{2}$, $\tau = 3 \Rightarrow M = 7$.

(I) $h_a(n) = \frac{\sin \frac{\pi}{2} (n-3)}{\pi (n-3)}$

: symmetric abt $n=3$.

$\therefore h_a(3) = \frac{1}{2}$

$h_a(0) = h_a(6) = \frac{1}{3\pi}$

$h_a(1) = h_a(5) = 0$

$h_a(2) = h_a(4) = \frac{1}{\pi}$

(II) $w_t(n) = \left\{ 0, \frac{1}{3}, \frac{2}{3}, 1, \frac{2}{3}, \frac{1}{3}, 0 \right\}$

(III) Impulse Response $h(n) = h_a(n) \cdot w_t(n)$

$= \left\{ 0, 0, \frac{2}{3\pi}, \frac{1}{2}, \frac{2}{3\pi}, 0, 0 \right\}$

(IV) Freq. Response $H(e^{j\omega}) = e^{-j3\omega} \left[\frac{1}{2} + \frac{4}{3\pi} \cos \omega \right]$

(V) System function $H(z) = \sum_{n=0}^6 h(n) z^{-n} = \frac{2}{3\pi} [z^{-2} + z^{-4}] + \frac{1}{2} z^{-3}$

(VI) Diff Eq. $y(n] = \frac{2}{3\pi} (x[n-2] + x[n-4]) + \frac{1}{2} x[n-3]$

